

NYC Taxi Fare Data

Automatidata Project — Executive Summary

Project Overview

22,699 NYC yellow taxi trips (2017) reviewed for structure, quality, and predictive potential ahead of fare-amount modelling.

Key Insights

- No missing values across all 22,699 trips and 18 columns
- Data quality issues found: negative fares/taxes, 0-passenger and 0-distance trips, invalid RatecodeID (99)
- A handful of extreme outliers (e.g., \$1,200 total_amount) skew the averages
- trip_distance and trip duration look like the strongest predictors of fare_amount

Details

VendorID	passenger_count	trip_distance
22699.000000	22699.000000	22699.000000
1.556236	1.642319	2.913313
0.496838	1.285231	3.653171
1.000000	0.000000	0.000000
1.000000	1.000000	0.990000
2.000000	1.000000	1.610000
2.000000	2.000000	3.060000
2.000000	6.000000	33.960000

Chart: fare_amount distribution, flagged outliers highlighted

Categorical fields (VendorID, RatecodeID, payment_type, zone IDs) shouldn't be averaged. Pickup/drop-off timestamps need conversion to date-time to engineer trip duration.

Next Steps

- Convert date-time columns; engineer trip_duration
- Filter or investigate negative and outlier records
- Confirm handling of 0-passenger and 0-distance trips
- Proceed to full EDA using the cleaned dataset